

FintechCo Payments Product

6-Month CAC Payback: Risk Analysis & Recommendations

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Executive Summary

Three months' worth of data are insufficient to substantiate a 6-month CAC payback, and this is the take-home message from the analysis. The most tenured group of customers (in the January group) still has only 89-90 days' worth of experience relative to a 180-day goal, and therefore, no customer has even made it to the point where payback was expected. All payback numbers cited hereafter are projections, not observations, and two projections were considered.

What the data shows

- CAC blended across all channels is \$60.78/client vs. \$133,414 in total Q1 acquisition spend for 2,195 new clients. Current average gross profit generated per client to date stands at \$9.53 (median \$3.16), representing approximately 16% of CAC recovered within the average tenure of ~37 days to date.
- Maturity has a big impact on client profitability: the average daily profit per active client increases nearly 15-20X from the first 10 days (~\$0.03/day) to the 70-89 days (~\$0.55-0.60/day) period. An average flat run-rate calculation will thus significantly understate the actual trend for mature clients, who can only be measured by the January cohort.
- Using a conservative flat run-rate estimate up to day 180, estimated CAC recovery will fall between 49% (Facebook) and 127% (Google) by sub-channel. Using a more optimistic run-rate estimate based on the most mature client group, the CAC recovery range moves up to 81% (LinkedIn) and 155% (Google).
- Two structural problems limit payback no matter which projection turns out to be true: 31.6% of the clients have been worth exactly \$0 in gross profit until now, and profit distribution is skewed (the top 10% of the clients earn 54% of all profits).
- Month of signing is another problem: matched tenure January clients perform slightly better than February clients, who perform better than March clients, an early indication of deteriorating quality of cohorts, although the latest cohort's sample size is quite small.

Bottom line for leadership management team

I suggest to treat 6 months of payback as unverified rather than achieved or missed, emphasizing three key steps for now:

- (1) Develop a live payback per cohort dashboard so that it doesn't become an ad hoc effort anymore,
- (2) Look into the 32% of customers who provide no contribution at all to see how structural impossibility of payback can happen when a third of all acquired customers don't pay back anything at all, and
- (3) Verify the premium CAC of FundCo (almost twice as high as any other sub-channel) through its better economics.

1. Data & Methodology

1.1 What's in the data

The source workbook contains two sheets:

- **Clients** : 84,313 daily records for 2,195 unique clients, from each client's account_created date through March 31, 2020. Columns: client ID, account_created date, channel (Direct / Partner), sub-channel (Facebook, Google, LinkedIn, AffilCo, FundCo, ReferCo), calendar date, and that day's gross profit.
- **Spend** : A small monthly table of acquisition spend by channel/sub-channel for January, February, and March (6 sub-channels $\times$ 3 months = 18 figures, \$133,414 total).

1.2 Data quality checks performed

Data Quality Check	Result
Missing / Null Values	None found in any column
Duplicate Records	No duplicate (client, activity_date) records
Negative Gross Profit	None found; minimum gross profit = \$0.00
Date Coverage	Every client has a continuous daily record from account_created through 2020-03-31
Channel Consistency	Each client is associated with exactly one channel and one sub-channel throughout the dataset
Outlier Check	Maximum single-day gross profit = \$7.32; highest cumulative client gross profit = \$138.08 (FundCo). No values indicate likely data-entry errors.

1.3 Assumptions made

The following items have not been mentioned explicitly in the source documents; rather, they have been included in this report because of the need to conduct an analysis.

Assumption	Assumption Made	Reason
Observation Cutoff	31 Mar 2020 is treated as the analysis (as-of) date for all tenure calculations.	The dataset does not specify an explicit analysis date.
Spend-to-Cohort Attribution	Marketing spend in each month and sub-channel is assumed to acquire the new clients from that same month and sub-channel.	The spend data is not linked to individual clients and provides no lag-adjusted attribution.
Gross Profit Definition	Gross profit is assumed to be net of all variable costs, as described in the assignment. No additional COGS adjustments were made.	No data dictionary or cost breakdown is provided.
Customer Churn	A day with \$0 gross profit is treated as an inactive day rather than customer churn or account closure.	The dataset contains no account status, closure, or churn information.

Customer Acquisition Cost (CAC)	The full reported marketing spend is treated as Customer Acquisition Cost (CAC). No allocation of fixed or overhead costs is applied.	The spend data does not distinguish between fixed and variable marketing costs.
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1.4 The core data limitation

The product was introduced 3 months prior to the current analysis, and the targeted payback period is 6 months (180 days). None of the customers within the data set has been in existence for 180 days the longest tenured customers (January customers) have only 89 to 90 days in the data set. The payback percentages used in the current analysis are all estimates made roughly 2x the size of the data set period.

2. Key Calculations

2.1 CAC by sub-channel

Channel	Sub-channel	New Clients (Q1)	Total Spend (Q1)	Blended CAC	Zero-Profit Rate
Direct	Facebook	531	\$11,486.74	\$21.63	44.8%
Direct	Google	306	\$7,673.67	\$25.08	35.3%
Direct	LinkedIn	141	\$3,167.05	\$22.46	29.8%
Partner	AffilCo	447	\$26,784.61	\$59.92	28.0%
Partner	FundCo	422	\$72,079.73	\$170.81	16.8%
Partner	ReferCo	348	\$12,222.00	\$35.12	31.3%

FundCo has the largest CAC (2.4x that of the next most costly sub-channel) yet the lowest zero-profit client percentage and the highest client average profit of any channel (\$28.13 compared to an average of \$9.53), the high investment seems to be paying off with higher quality clients. Facebook is exactly the opposite - having the smallest CAC in the Direct category but the highest number of zero-profit clients (44.8%).

2.2 Profit distribution & concentration

- Overall average gross profit to date: \$9.53/client; median: \$3.16 the large gap between mean and median indicates a right-skewed distribution driven by a minority of high-value clients.
- 31.6% of all clients (694 of 2,195) have generated exactly \$0 in gross profit to date.
- The top 10% of clients (by profit) account for 53.8% of all gross profit generated; the top 1% alone account for 10.3%.

2.3 Profitability ramp curve

Combining all customers based on their days since acquisition, the average gross profit per day for each active customer increases constantly throughout the product life cycle:

Days Since Acquisition	Average Gross Profit per Active Client-Day
0-9	\$0.03
10-19	\$0.16
20-29	\$0.24
30-39	\$0.34
40-49	\$0.41
50-59	\$0.44
60-69	\$0.52
70-79	\$0.62
80-89	\$0.58

This follows the logic of a usage ramp approach; as users become familiar with the product, they interact with it more. It is also important to mention that, by definition, a flat forecast for the entire period (Scenario A below) is conservative. Yet all the data points from ~day 60 onwards are derived almost exclusively from the January cohort as it is the only one sufficiently mature enough to be at such a tenure level.

2.4 Cohort comparison (quality-drift check)

At matched tenure (same days-since-acquisition), average cumulative profit by acquisition cohort:

Days Since Acquisition	January Cohort Avg.	February Cohort Avg.	March Cohort Avg.
10	\$0.49	\$0.52	\$0.43
20	\$2.41	\$2.27	\$1.95
30	\$5.14	\$4.68	\$3.71 (<i>n = 39 clients</i>)
45	\$10.90	\$10.15	-
55	\$15.25	\$13.73	-

2.5 Payback projections to day 180

Two extrapolation methods, both explained in full in the companion workbook's Payback_Projection:

- **Scenario A (conservative):** The entire lifetime average revenue per day for each sub-channel is assumed to be flat and is multiplied by 180 days. This scenario underestimates the real trajectory because it does not consider the curve in 2.3.
- **Scenario B (optimistic):** The most mature run-rate for any observed client (days 70-89, January clients only) is held flat and is multiplied by 180 days. This scenario overestimates the trajectory because it assumes that all future clients will ultimately behave like mature January clients (2.4).

Sub-channel	CAC	Scenario A: CAC Recovered by Day 180	Scenario B: CAC Recovered by Day 180
Facebook	\$21.63	49%	89%
Google	\$25.08	127%	155%
LinkedIn	\$22.46	98%	81%
AffilCo	\$59.92	58%	127%
FundCo	\$170.81	62%	135%
ReferCo	\$35.12	91%	95%

The true outcome likely sits between these two scenarios. Treat the gap between them not either single number as the honest measure of uncertainty. In **Scenario A**, only Google attains the 100% mark on Day 180, but in **Scenario B**, Google, AffilCo, and FundCo reach this mark while Facebook and LinkedIn still remain unclear.

3. Risk Analysis

Ranked by combined likelihood and potential impact on the 6-month payback goal.

Severity	Risk	Business Impact
High	No client has reached the 180-day payback horizon	Any 180-day CAC payback estimate is based on extrapolation beyond the observed data (maximum observed tenure $\approx$ 90 days). Decisions to scale or reduce marketing spend may therefore be based on uncertain assumptions.
High	Projection methodology significantly affects results	Conservative and optimistic projection methods estimate CAC recovery between 49% and 155% for the same sub-channels. Small modeling assumptions can change the conclusion from unprofitable to profitable, so no single projection should be treated as definitive.
High	31.6% of clients generate zero gross profit	Nearly one-third of acquired customers contribute nothing toward CAC recovery, limiting overall payback until these customers activate or acquisition targeting improves.
Medium-High	Channel concentration and early cohort quality drift	FundCo accounts for approximately 54% of total marketing spend, making overall profitability highly dependent on its performance. February and March cohorts also show slightly weaker early profitability than the January cohort.
Medium	Profit is concentrated among a small group of customers	The top 10% of customers generate 54% of total gross profit. Losing a small number of high-value customers could materially reduce overall profitability.
Medium	Limited sample size at longer customer tenures	Customers with 80+ days of tenure are few and belong almost entirely to the January cohort, increasing statistical uncertainty in long-term profit projections.
Medium	No customer retention or churn information	Without churn or account-status data, it is impossible to distinguish inactive customers from churned customers, limiting the accuracy of lifetime value (LTV) and payback analyses.
Low-Medium	Rapid customer acquisition creates an immature customer base	Marketing spend and customer acquisition increased each month, resulting in many recently acquired customers. Current blended performance metrics may therefore understate long-term profitability as cohorts mature.

4. Recommendations

4.1 Immediate actions (next 30 days)

Recommended Action	Expected Impact	Priority
Build a live cohort payback dashboard (tracking cumulative profit by days since acquisition and channel with automatic refresh)	Enables continuous monitoring of CAC recovery, eliminates repetitive manual analysis, and makes projection uncertainty transparent for decision-makers.	High
Analyze the 694 zero-profit customers by separating never-activated users from onboarded-but-inactive users	Nearly one-third of customers generate no gross profit. Improving activation rates could significantly increase overall CAC recovery and channel profitability.	High
Validate FundCo's high CAC against recent unit economics before increasing marketing spend	FundCo accounts for 54% of total marketing spend and has the highest CAC. Confirm that recent cohorts continue to justify the premium acquisition cost before further investment.	High
Limit additional Facebook marketing spend until the causes of its high zero-profit rate are identified	Facebook shows the highest zero-profit rate (44.8%) and the weakest projected CAC recovery. Investigating targeting, onboarding, or campaign quality before scaling spend can reduce acquisition risk.	Medium

4.2 Medium-term actions (next 90 days)

Recommended Action	Expected Impact	Priority
Extend the observation window and rebuild the CAC payback analysis using actual 150-180 day customer data	Replaces projected estimates with observed performance, reducing uncertainty in 180-day payback calculations and validating the current projection models.	High
Conduct a controlled budget reallocation experiment by shifting incremental spend from Facebook/AffilCo to Google/FundCo	Google and FundCo demonstrate the strongest projected unit economics. A small-scale experiment can validate performance before committing to a larger budget reallocation.	Medium
Investigate the declining cohort performance from January to February to March	Determine whether the trend is driven by changes in targeting, market conditions, seasonality, or data quality. Persistent deterioration would reduce confidence in future payback forecasts.	Medium
Add customer retention and churn tracking to the client dataset	Enables accurate customer lifetime value (LTV), retention, and attrition analysis instead of relying solely on cumulative gross profit as a proxy for customer value.	Medium

4.3 Long-term monitoring

Recommended Action	Expected Impact	Priority
Establish a recurring CAC payback dashboard by channel and cohort, refreshed monthly using actual customer performance	Converts this one-time analysis into an ongoing performance management process and tracks actual CAC recovery as cohorts mature beyond 180 days.	High
Implement channel-level marketing spend gates based on observed unit economics and review them quarterly	Prevents marketing budgets from being scaled based solely on optimistic projections and ensures decisions are driven by validated performance data.	Medium
Monitor profit concentration and develop engagement initiatives for lower-value customers	Reduces dependence on a small group of high-value customers and improves the profitability of the broader customer base over time.	Low-Medium

5. Key Assumptions and Limitations

5.1 Monthly Spend-to-Customer Attribution

The analysis assumes that marketing spend in a particular month and sub-channel resulted in customers acquired during the same month. Since the dataset does not provide customer-level attribution or conversion lag information, this assumption was necessary to calculate CAC. Actual acquisition timing may vary, causing slight differences in channel-level CAC estimates.

5.2 Projection Based on Available Customer History

The dataset contains only three months of observations, so January customers are the only cohort with sufficient history for forecasting. The six-month projection assumes that February and March customers will follow a similar profitability trend as they mature. Actual performance may vary as more data becomes available.

5.3 Zero-Profit Customers

Customers with zero gross profit were treated as inactive because the dataset does not include information on churn, activation status, or transaction failures. Therefore, zero profit should not be interpreted as customer loss, but rather as a limitation of the available data.

6. Conclusion

Based on the data and analysis presented in this report, CAC recovery within six months cannot yet be confirmed. The dataset covers only the first three months since launch, and even the most tenured client group, the January cohort, has just 89-90 days of history against the 180-day target. No client has actually reached the payback horizon, so every recovery figure in this report is a projection rather than an observed outcome, and two projection methods were used precisely to reflect that uncertainty rather than present a single, false-certain number.

The projections themselves show a wide range: a conservative flat run-rate estimate puts CAC recovery by day 180 between 49% (Facebook) and 127% (Google), while a more optimistic estimate based on the most mature clients raises that range to 81% (LinkedIn) and 155% (Google). **Google** therefore appears the strongest candidate for full CAC recovery under either method, while **Facebook** is the weakest performer under the conservative scenario. **FundCo's** acquisition cost is markedly higher than other sub-channels and warrants direct verification against its underlying economics before any conclusion is drawn on its payback prospects.

Beyond channel-level performance, two structural risks limit payback regardless of which projection proves correct: 31.6% of clients have generated zero gross profit to date, and profit is heavily concentrated, with the top 10% of clients responsible for 54% of all profit generated. A further risk is early evidence of declining cohort quality, with January clients outperforming February clients, who in turn outperform March clients at matched tenure.

Given these findings, **the recommended path forward is to treat 6-month payback as unverified rather than achieved or missed, and to prioritize three actions:**

- 1) building a live, cohort-based payback dashboard**
- 2) investigating the segment of clients contributing no profit and**
- 3) validating FundCo's higher acquisition cost against its actual return before any further conclusions are drawn.**